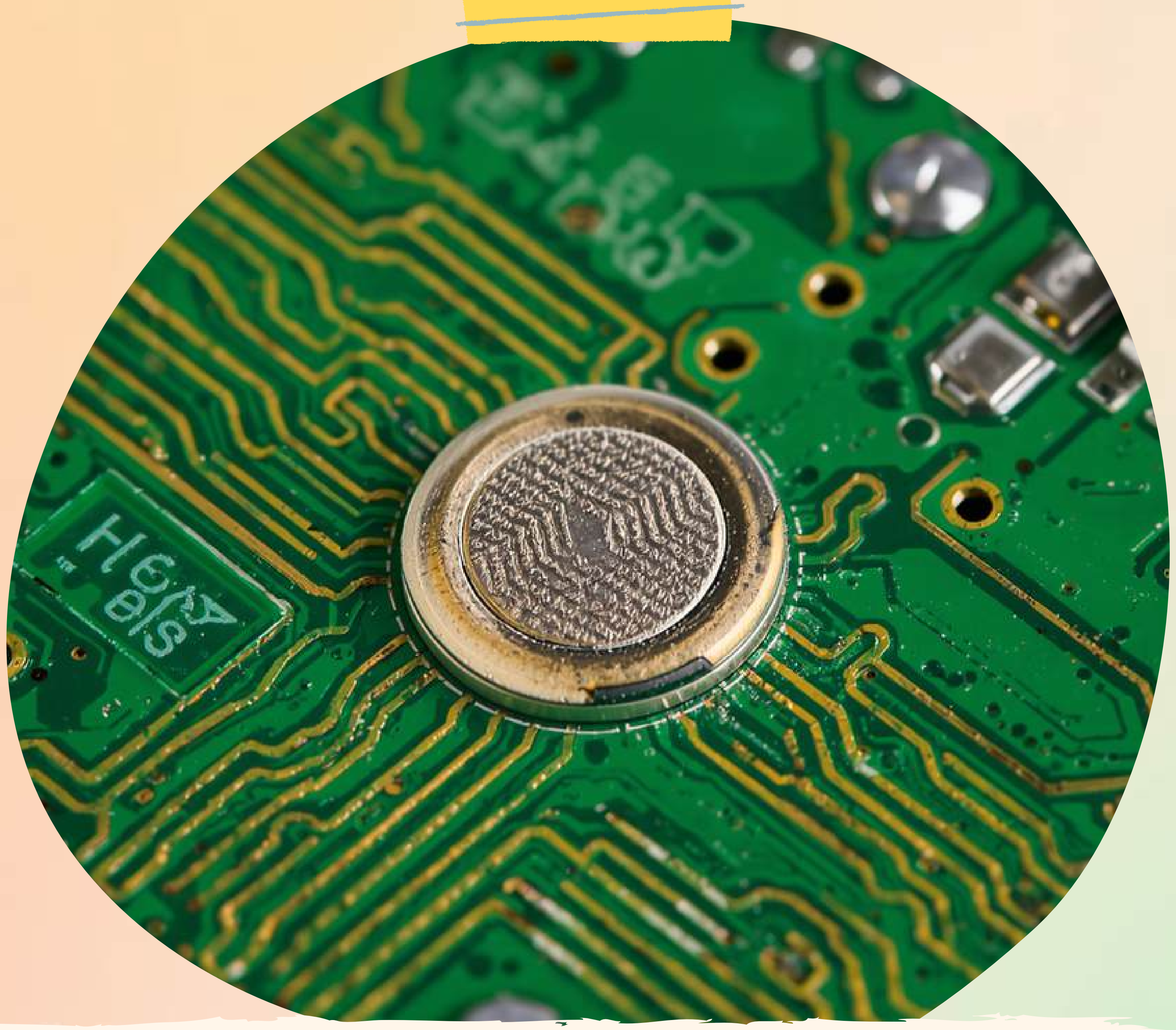
LIGHT-DEPENDENT RESISTOR (LDR)

HOW DO STREETLIGHTS KNOW WHEN TO GLOW?

A SENSOR IS LIKE MINI-EYES FOR ELECTRONICS!

SENSOR

It helps things 'see' the world around them!
Example: Delhi Metro's automatic doors open by sensing people nearby — just like a pair of invisible eyes watching and reacting!



MEET THE LDR — DARKNESS DETECTIVE!

HOW DOES LDR WORK? — THE WAVY LINE STORY

ALERT!

When bright light waves hit LDR, it relaxes. 😎☀️
When darkness covers it, LDR becomes alert! 🌙
Light = Low resistance. Dark = High resistance!
That's how LDR decides when to switch lights ON!

INTERACTIVE CHECKPOINT 1

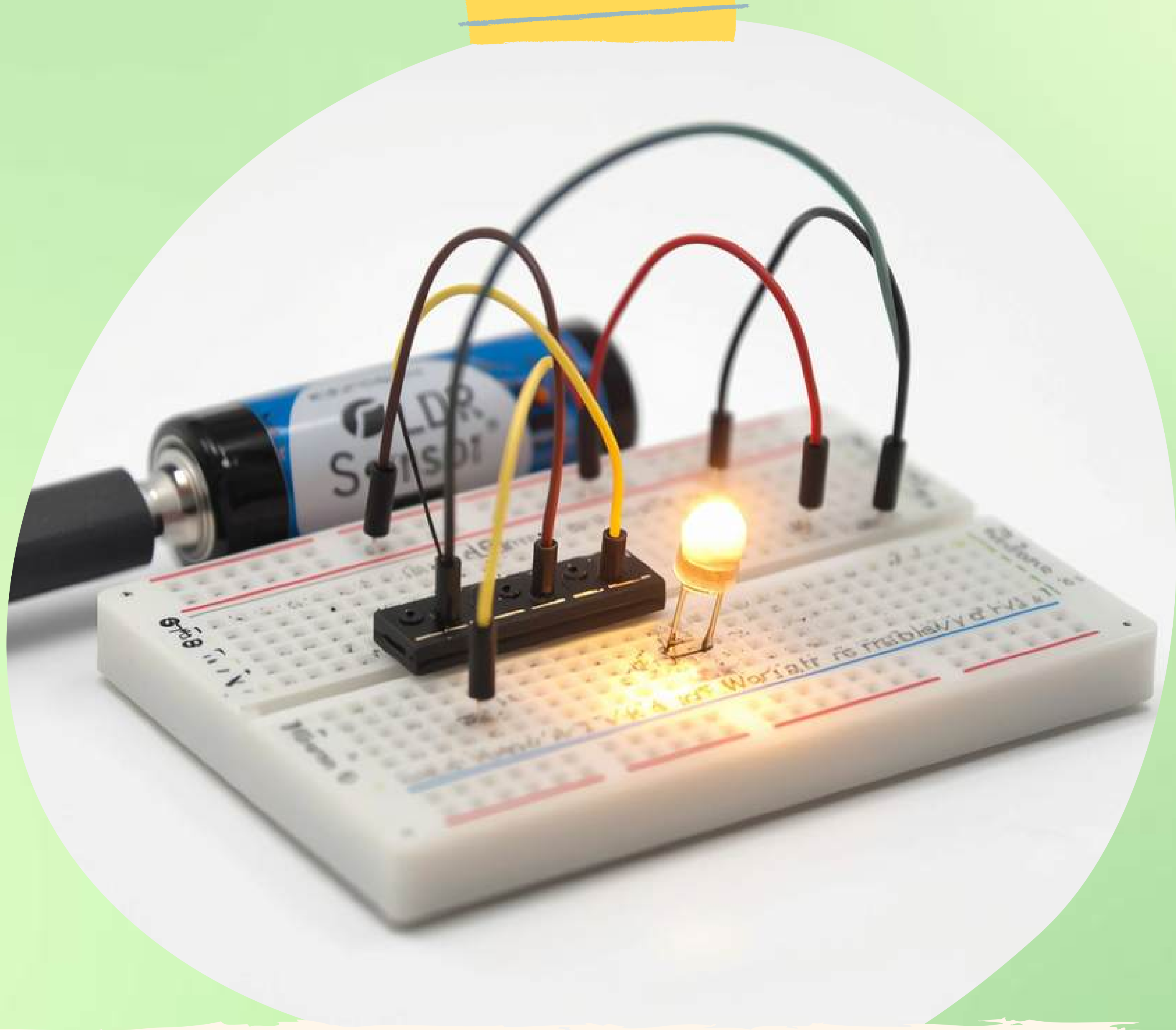
WHAT WILL THE LDR DO WHEN NIGHT FALLS AND IT GETS DARK?

A. RELAX

B. GET SUPER ACTIVE

C. SLEEP

D. MAKE NOISE

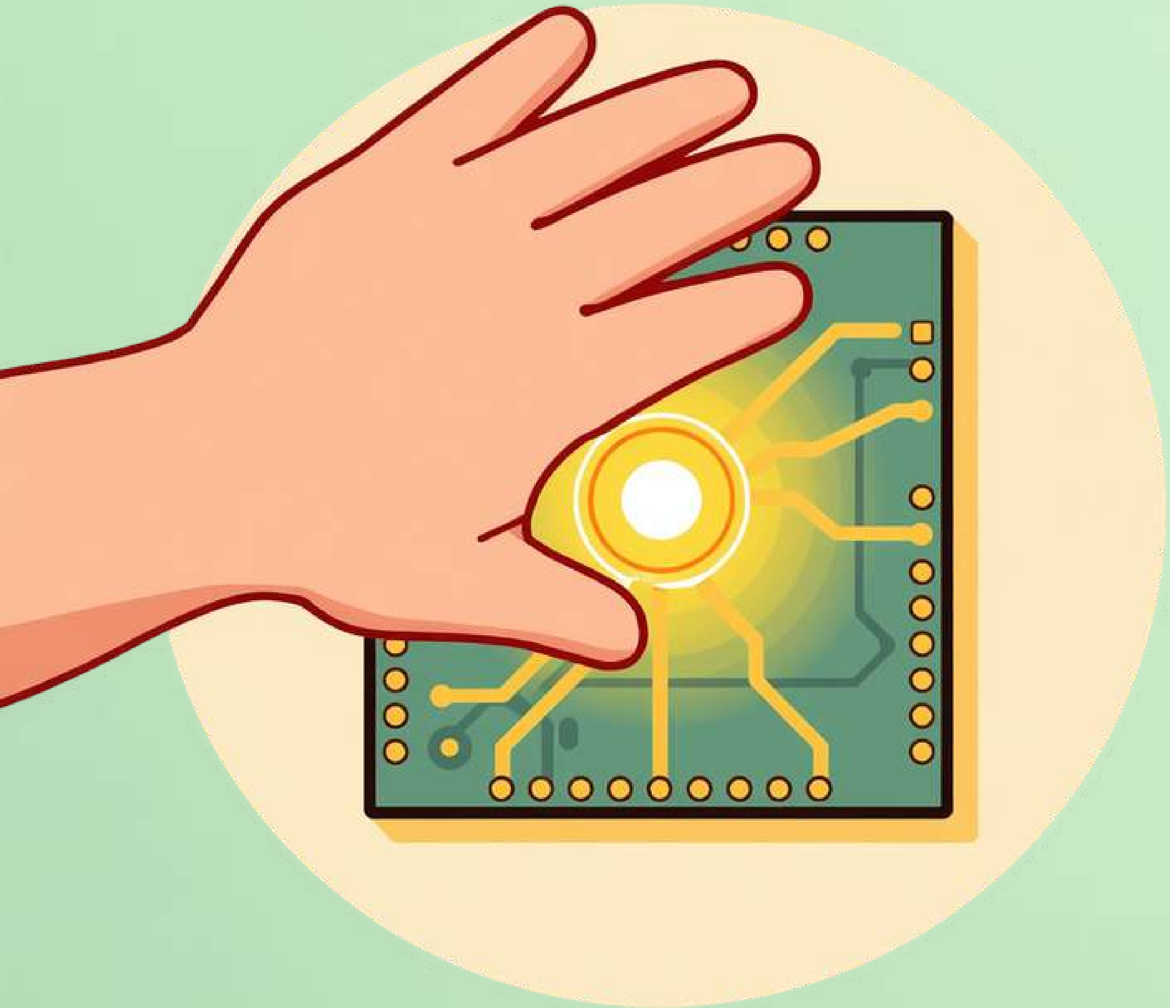


THE AUTOMATIC NIGHT LIGHT CIRCUIT

WATCH THE LED LIGHT TURN ON WHEN THE

SUN SETS!

In our virtual circuit, move the sun down and watch the LED glow bright!
When the sun rises, the LED turns OFF.
The LDR sees the change — just like magic!



INTERACTIVE CHECKPOINT 2

IF WE COVER THE LDR SENSOR WITH OUR HAND, WHAT HAPPENS TO THE LED LIGHT?

A. TURNS ON

B. TURNS OFF

C. BLINKS

D. NOTHING HAPPENS

REAL-WORLD CONNECTIONS IN INDIA

SOLAR STREETLIGHTS

SMART CITY LIGHTS

RAILWAY COACH LIGHTS

DIWALI NIGHT BULBS

SAVE POWER!

QUIZ TIME!

WHAT DOES A SENSOR DO?

A. MINI-EYES FOR ELECTRONICS

B. EARS FOR LISTENING

C. HANDS FOR TOUCHING

D. NOSE FOR SMELLING

QUIZ TIME! — QUESTION 2

WHEN DOES THE LDR GET SUPER ACTIVE?

A. BRIGHT LIGHT

B. DARKNESS

C. RAIN

D. WIND

QUIZ TIME!

Q3: NAME ONE PLACE IN INDIA WHERE AUTOMATIC LIGHTS USE LDR.

A. METRO STATION

B. VILLAGE STREETLIGHT

C. SCHOOL CLASSROOM

D. SHOPPING MALL

SUMMARY

WHAT WE DISCOVERED TODAY!

SENSORS ARE MINI-EYES FOR ELECTRONICS!

LDR IS THE DARKNESS DETECTIVE — ON IN DARK, OFF IN LIGHT!

AUTOMATIC LIGHTS SAVE POWER EVERY DAY!

COVER LDR = LIGHT ON. BRIGHT LIGHT = LIGHT OFF!

Thank you for learning with us!

ANY QUESTIONS?